



```
In [16]: import nltk
import spacy
import matplotlib.pyplot as plt
import pandas as pd
# Download NLTK data
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
# Download spaCy English model
# If this fails, try: !python -m spacy download en_core_web_sm
try:
    nlp = spacy.load('en_core_web_sm')
except OSError:
    print('Downloading spaCy model (en_core_web_sm)...')
    !python -m spacy download en_core_web_sm
    nlp = spacy.load('en_core_web_sm')

# Define the academic essay text
essay_text = """This essay investigates the complex interplay between technolo
print("Libraries imported, NLTK data and spaCy model downloaded, and essay tex
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]   /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]   date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]   date!
```

Libraries imported, NLTK data and spaCy model downloaded, and essay text defined.

```
In [17]: tokens_nltk = nltk.word_tokenize(essay_text)
pos_tags_nltk = nltk.pos_tag(tokens_nltk)
print("NLTK POS tagging complete. First 10 tags:")
print(pos_tags_nltk[:10])
```

NLTK POS tagging complete. First 10 tags:

```
[('This', 'DT'), ('essay', 'NN'), ('investigates', 'VBZ'), ('the', 'DT'), ('com-
plex', 'JJ'), ('interplay', 'NN'), ('between', 'IN'), ('technological', 'JJ'),
('advancements', 'NNS'), ('and', 'CC')]
```

```
In [18]: doc_spacy = nlp(essay_text)
pos_tags_spacy = [(token.text, token.pos_) for token in doc_spacy]
print("spaCy POS tagging complete. First 10 tags:")
print(pos_tags_spacy[:10])
```

spaCy POS tagging complete. First 10 tags:

```
[('This', 'DET'), ('essay', 'NOUN'), ('investigates', 'VERB'), ('the', 'DET'), ('complex', 'ADJ'), ('interplay', 'NOUN'), ('between', 'ADP'), ('technological', 'ADJ'), ('advancements', 'NOUN'), ('and', 'CCONJ')]
```

```
In [19]: nltk_nouns = [word for word, tag in pos_tags_nltk if tag.startswith(('NN', 'NNP'))]
nltk_verbs = [word for word, tag in pos_tags_nltk if tag.startswith(('VB', 'VBD', 'VBN', 'VBP', 'VBZ'))]
spacy_nouns = [token for token, tag in pos_tags_spacy if tag in ('NOUN', 'PROPN')]
spacy_verbs = [token for token, tag in pos_tags_spacy if tag in ('VERB', 'AUX')]
print("NLTK Nouns (first 10):", nltk_nouns[:10])
print("NLTK Verbs (first 10):", nltk_verbs[:10])
print("spaCy Nouns (first 10):", spacy_nouns[:10])
print("spaCy Verbs (first 10):", spacy_verbs[:10])
```

NLTK Nouns (first 10): ['essay', 'interplay', 'advancements', 'structures', 'innovations', 'intelligence', 'instance', 'labor', 'markets', 'concerns']

NLTK Verbs (first 10): ['investigates', 'have', 'altered', 'raised', 'regarding', 'necessitates', 'adapt', 'prepare', 'dominated', 'requires']

spaCy Nouns (first 10): ['essay', 'interplay', 'advancements', 'structures', 'innovations', 'intelligence', 'instance', 'labor', 'markets', 'concerns']

spaCy Verbs (first 10): ['investigates', 'have', 'altered', 'raised', 'regarding', 'necessitates', 'must', 'adapt', 'prepare', 'dominated']

```
In [20]: from collections import Counter
# NLTK Nouns Frequency
nltk_noun_freq = Counter(nltk_nouns)
df_nltk_nouns = pd.DataFrame(nltk_noun_freq.most_common(), columns=['Word', 'Frequency'])
# NLTK Verbs Frequency
nltk_verb_freq = Counter(nltk_verbs)
df_nltk_verbs = pd.DataFrame(nltk_verb_freq.most_common(), columns=['Word', 'Frequency'])
# spaCy Nouns Frequency
spacy_noun_freq = Counter(spacy_nouns)
df_spacy_nouns = pd.DataFrame(spacy_noun_freq.most_common(), columns=['Word', 'Frequency'])
# spaCy Verbs Frequency
spacy_verb_freq = Counter(spacy_verbs)
df_spacy_verbs = pd.DataFrame(spacy_verb_freq.most_common(), columns=['Word', 'Frequency'])
print("--- NLTK Nouns Frequency (Top 10) ---")
print(df_nltk_nouns.head(10))
print("\n--- NLTK Verbs Frequency (Top 10) ---")
print(df_nltk_verbs.head(10))
print("\n--- spaCy Nouns Frequency (Top 10) ---")
print(df_spacy_nouns.head(10))
print("\n--- spaCy Verbs Frequency (Top 10) ---")
print(df_spacy_verbs.head(10))
```

--- NLTK Nouns Frequency (Top 10) ---

	Word	Frequency
0	essay	1
1	interplay	1
2	advancements	1
3	structures	1
4	innovations	1
5	intelligence	1
6	instance	1
7	labor	1
8	markets	1
9	concerns	1

--- NLTK Verbs Frequency (Top 10) ---

	Word	Frequency
0	are	2
1	investigates	1
2	have	1
3	altered	1
4	raised	1
5	regarding	1
6	necessitates	1
7	adapt	1
8	prepare	1
9	dominated	1

--- spaCy Nouns Frequency (Top 10) ---

	Word	Frequency
0	essay	1
1	interplay	1
2	advancements	1
3	structures	1
4	innovations	1
5	intelligence	1
6	instance	1
7	labor	1
8	markets	1
9	concerns	1

--- spaCy Verbs Frequency (Top 10) ---

	Word	Frequency
0	must	2
1	are	2
2	investigates	1
3	have	1
4	altered	1
5	raised	1
6	regarding	1
7	necessitates	1
8	adapt	1
9	prepare	1

In [21]: `import matplotlib.pyplot as plt`

```

top_n = 10

# Plot NLTK Nouns
plt.figure(figsize=(10, 6))
plt.bar(df_nltk_nouns['Word'].head(top_n), df_nltk_nouns['Frequency'].head(top_n))
plt.title(f'Top {top_n} NLTK Noun Frequencies')
plt.xlabel('Nouns')
plt.ylabel('Frequency')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

# Plot NLTK Verbs
plt.figure(figsize=(10, 6))
plt.bar(df_nltk_verbs['Word'].head(top_n), df_nltk_verbs['Frequency'].head(top_n))
plt.title(f'Top {top_n} NLTK Verb Frequencies')
plt.xlabel('Verbs')
plt.ylabel('Frequency')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

# Plot spaCy Nouns
plt.figure(figsize=(10, 6))
plt.bar(df_spacy_nouns['Word'].head(top_n), df_spacy_nouns['Frequency'].head(top_n))
plt.title(f'Top {top_n} spaCy Noun Frequencies')
plt.xlabel('Nouns')
plt.ylabel('Frequency')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

# Plot spaCy Verbs
plt.figure(figsize=(10, 6))
plt.bar(df_spacy_verbs['Word'].head(top_n), df_spacy_verbs['Frequency'].head(top_n))
plt.title(f'Top {top_n} spaCy Verb Frequencies')
plt.xlabel('Verbs')
plt.ylabel('Frequency')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

```



